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Brief Report

# Evaluating the Factor Structure of the Preliminary Version of EuroQol Health and Well-Being Instrument in China: A Replication of the Confirmatory Factor Analysis



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## ABSTRACT

**Objectives:** During the development of EuroQol Health and Well-Being instrument (EQ-HWB), the Chinese online sample showed a weaker model fit for the confirmatory factor model based on the UK sample, even after adjustments. This may be because of cultural differences affecting item interpretation or demographic disparities, particularly in age and education. To address this, the study aimed to replicate the Chinese-adjusted model, a version of the UK model modified based on the Chinese online sample, using a more diverse Chinese face-to-face sample to assess potential improvements in model fit.

**Methods:** We conducted face-to-face interviews to recruit respondents with diverse age and education backgrounds. The Chinese-adjusted model, a bifactor confirmatory model with 9 factors and 2 measurement factors, was replicated in 2 stages. First, the Chinese online sample validated the model. Then, the face-to-face sample assessed improvements in model fit. The confirmatory factor analysis model fit criteria were as follows: comparative fit index and Tucker-Lewis index greater than 0.95 and root mean square error of approximation less than 0.06.

**Results:** We recruited 553 respondents, including 96 with lung cancer, 125 with diabetes, 102 with depression, 113 with schizophrenia, 16 with other diseases, and 101 healthy individuals. In the face-to-face sample, the goodness-of-fit indices for the Chinese-adjusted model were a comparative fit index of 0.943, Tucker-Lewis index of 0.939, and root mean square error of approximation of 0.052 (90% CI 0.050–0.054), indicating a significantly better fit than in the original study.

**Conclusions:** The misfit of the UK EQ-HWB model to Chinese data in a previous study had seemed to be because of limited variance. Using a more diverse data set, our study supported the EQ-HWB factor structure's comparability at an international level.

**Keywords:** confirmatory factor analysis, EQ-HWB, health and well-being, measurement properties.

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## Introduction

Quality-adjusted life-years (QALYs), which incorporate health-related quality of life, are commonly used to quantify the benefits of interventions in economic evaluations.<sup>1,2</sup> Standard health-related quality of life instruments, such as the EQ-5D and the Short-Form 6 Dimensions, provide the necessary valuation sets for calculating QALYs.<sup>3,4</sup> Nevertheless, these instruments have been criticized for their limited scope in capturing other benefits of interventions that are not directly related to health, such as autonomy and cognition. In addition, they tend to focus primarily on outcomes within healthcare systems. Therefore, the “Going beyond health-related quality of life – towards a broader QALY measure for use across sectors” (E-QALY) project was launched to design a generic instrument, which can capture quality of life (QoL) related to both healthcare and social care sectors.<sup>5</sup> The resulting EuroQol Health and Well-Being (EQ-HWB)

is developed as a new generic instrument, measures health and well-being outcomes related to interventions among healthy individuals, patients, and caregivers across healthcare and social care sectors.<sup>6–9</sup> To ensure the EQ-HWB's international applicability, respondents from the United Kingdom, United States, Germany, China, Australia, and Argentina were recruited during its development stages.<sup>10</sup>

During the psychometric survey stage, the analysis results demonstrated consistency across 6 countries, except for the performance of the confirmatory factor analysis (CFA) in the Chinese population. CFA was used to explore the conceptual framework and support decision making regarding the contents and domains of the EQ-HWB.<sup>10</sup> Peasgood et al<sup>10</sup> applied a bifactor model to account for items with both positive and negative phrasing. The CFA model was constructed using the UK sample, which included data from an online panel and offline patient samples. The 64 candidate items were categorized into 7 domains

and subdomains, namely feelings and emotions (happiness, hope, anxiety, safety, anger), activity (self-care, enjoyable activity, mobility), autonomy (control, coping), physical sensations (pain, energy), self-respect (self-worth, self-respect), and relationships (loneliness, support, stigma, belonging, positive relationships).<sup>10</sup>

The UK bifactor CFA model was developed to represent the intended conceptual framework of the EQ-HWB. To assess its generalizability, the model was tested in five additional countries, including Argentina, Germany, the United States, Australia, and China. Although the original UK model fit well for the United States and Australian samples, modifications were needed in Argentina, Germany, and China to achieve an acceptable fit. In the German model, mobility and self-care were combined, and the energy facet was removed. The Argentinean model applied similar changes, merging mobility with self-care, removing the energy facet, and integrating self-worth into the coping factor. The Chinese model required more extensive modifications, including merging mobility with self-care, consolidating anxiety, happiness, hope, coping, and self-worth into a single factor, and removing the energy facet. In addition, some items in the Chinese model did not align with the positive or negative measurement factors. After these adjustments, the Argentinean and German models achieved a good fit, whereas the Chinese model continued to present issues.<sup>10</sup>

Several factors can contribute to variations in factor fit. For example, populations with different ages or health conditions may exhibit different response patterns. Younger individuals tend to report more mental health issues, whereas older adults more frequently report physical health problems.<sup>11,12</sup> These differences in response patterns affect item intercorrelations, which can lead to items loading onto different factors or loading onto the same factors with varying strengths. Alternatively, cultural factors influence how respondents interpret and prefer response options, potentially affecting response patterns and challenging the direct comparability of factor structures at the international level. For instance, the exploratory factor structure of the EQ-HWB has been reported as 4 factors in an Australian sample but 5 factors in a Chinese sample.<sup>9,13</sup> These differences may arise from sample demographic variations, cultural influences, or measurement adaptations, all of which can impact the final factor structure.

The discrepancy in sample characteristics—particularly health conditions, age, and education—between the Chinese online sample and the UK sample was significant. The Chinese online sample had a skewed distribution, with only 5.2% of respondents over 50 years old and 91.2% holding undergraduate or postgraduate degrees.<sup>10</sup> This led to the hypothesis that the weak model fit may have been influenced by these demographic differences. To test this, this brief report aimed to replicate the CFA of the EQ-HWB conducted in the international psychometric study, using a new Chinese sample that better matched the UK sample in age and education. To enhance comparability, face-to-face interviews were conducted to recruit respondents with a broader range of ages and education levels.

## Methods

### Data Collection

To ensure consistency with the previous study, we used the existing questionnaire from the E-QALY. We recruited and trained 14 undergraduate students as interviewers. All interviewers received standardized data collection training on the background of this study and the introduction of the instruments.

According to Myers et al,<sup>14</sup> a commonly used rule for determining an adequate sample size for a CFA model is “ $N/P \geq 10$ ” (where  $N$  is the sample size and  $P$  is the number of variables in the model). Although the survey included 64 items, not all were used in constructing the CFA model. A sample size exceeding 500 was, therefore, considered reasonable. In addition, the original study's Chinese online sample also had a sample size of 500. To maintain consistency, we aimed for a similar sample size in this study.

Following the E-QALY protocol, this survey targeted 5 groups: 2 physical condition groups, 2 mental condition groups, and 1 healthy group. For the physical condition groups, a chronic illness and a common cancer were selected based on their high prevalence and significant disease burden in China. A similar approach was used when selecting the mental condition groups. As a result, the target sample included 100 patients with lung cancer, 100 with diabetes, 100 with depression, 100 with schizophrenia, and 100 from the general public. Patients were primarily recruited from local hospitals, whereas the general public was recruited through personal invitations. Face-to-face data collection took place from April to July 2021, whereas online data collection was conducted in 2019.

Each respondent provided oral informed consent before participating in the survey, and ethical approval was granted by Guizhou Medical University (approval letter number 2021-101). Respondents completed the survey following the same protocol as the original online survey, reporting their demographic status, socioeconomic characteristics, disease history, and caregiving experience.<sup>10</sup> All questionnaires were printed on an article and distributed to participants. Respondents were encouraged to respond to the questionnaire by themselves, and interviewers were able to explain the questions if necessary. Notably, respondents were instructed not to respond if they could not understand the item or felt that the item was irrelevant. After completing the survey, respondents received a small gift (valued at 50 Chinese Yuan) as a token of appreciation. All participants were recruited in Guizhou Province, China. Data were double-entered by the interviewers using EpiData.<sup>15</sup>

### Data Analysis

We followed the same steps to construct the Chinese-adjusted model, first fitting it to the original online sample data to ensure the model could be replicated. We then applied the newly collected face-to-face sample to test the Chinese-adjusted model. Notably, all 64 items were included in the questionnaires, but not all were incorporated into the factor model. Certain items related to sleep, vision, seeing, and discomfort were reported as unidimensional,<sup>10,16</sup> preventing their inclusion. Since the item pool contained both positively and negatively phrased items, 2 general measurement factors were defined, with positive and negative items loading onto separate factors.<sup>17</sup> Table 1 illustrates the structure of the factor model, showing the factor names and indicating which items load onto each factor.

The factors were freely estimated except for the 2 general factors (positive and negative factors), which were modeled as uncorrelated with the other factors. Missing data were handled using pairwise deletion under the weighted least squares mean and variance adjusted estimator, ensuring that cases with missing values were not completely excluded from the analysis.<sup>18</sup> The model goodness-of-fit indices were assessed using the criteria provided by Hu and Bentler: comparative fit index (CFI) > 0.95, Tucker-Lewis index (TLI) > 0.95, and root mean square error of approximation (RMSEA) < 0.06.<sup>19</sup> Because the response scale of all the items is a Likert-type scale, we used a

**Table 1.** Factor structure adjusted under the Chinese online sample.

| Domain       | Item          |                |           |              |                |            |             |            |                    |
|--------------|---------------|----------------|-----------|--------------|----------------|------------|-------------|------------|--------------------|
| Activity     | Do need       | Do value       | Enjoyed   | Do wanted    |                |            |             |            |                    |
| Self-care    | Need met      | Difficult wash | Personal  | Look after   | EQ5D self-care | Go outside | Activity    | Go outside | EQ-5D mobility     |
| Relationship | Got along     | Support need   | Accept    | Isolated     | Left out       | Close      | Lonely      | Avoid      | Exclude            |
|              | Unsupported   | Talk           |           |              |                |            |             |            |                    |
| Control      | No Control    | Able           | Control   | SWEMWBS mind |                |            |             |            |                    |
| Cognition    | Concentration | Remember       | Confused  | Think        | Attention      |            |             |            |                    |
| Anger        | Temper        | Frustration    | Irritable | Angry        |                |            |             |            |                    |
| Safety       | SAFE          | Afraid         | Unsafe    | Fright       |                |            |             |            |                    |
| Anxiety      | Calm          | Worried        | Anxious   | Happy        | Cheerful       | Sad        | Unhappy     | Hopeful    | Look forward       |
|              | Worth living  | Good           | Confide   | Unsure       | Failure        | Able cope  | Unable cope | Overwhelm  | SWEMWBS optimistic |

*Note.* Do need: I was able to do what I needed; Do value: I was able to do the things I value; Enjoyed: I enjoyed what I did; Do wanted: I could do the things I wanted to do; Need met: My personal needs were met (e.g. being washed, going to the toilet, getting dressed, having food when I needed); Difficult wash: How difficult is it for you to wash, toilet, dress yourself, eat or care for your appearance? Personal: How well were your personal needs met (e.g., being washed, going to the toilet, getting dressed, having food when you needed)?; Look after: I was able to look after myself (e.g., being washed, going to the toilet, getting dressed, having food when I needed); EQ-5D self-care: I have no problems washing or dressing myself; Go outside: How well were you able to get around outside?; Activity: How well were you able to do your day-to-day activities (e.g., working, shopping, traveling)?; Inside: How well were you able to get around inside your home?; EQ-5D mobility: I have no problems in walking about; Got along: I got along well with people around me; Support need: I had support when I needed it; Accept: I felt accepted by others; Isolated: I felt isolated; Left out: I felt left out; Close: I felt there was nobody I was close to; Lonely: I felt lonely; Avoid: I felt people avoided me; Exclude: I felt excluded; Unsupported: I felt unsupported by people; Talk: I felt I had no one to talk to; No control: I felt I had no control over my day-to-day life; Able: I was able to do the things I wanted to do; Control: I felt in control of my day-to-day life, By 'control over day-to-day life' we mean having the choice to do things or have things done for you as you like and when you want; SWEMWBS mind: I've been able to make up my own mind about things; Concentrate: I found it hard to concentrate; Remember: I had trouble remembering; Confused: I felt confused; Think: I had trouble thinking clearly; Attention: I found it hard to pay attention; Temper: I lost my temper easily; Frustration: I felt frustrated; Irritable: I felt irritable; Angry: I felt angry; Safe: I felt safe; Afraid: I felt afraid; Unsafe: I felt unsafe; Fright: I felt frightened; Calm: I felt calm; Worried: I felt worried; Anxious: I felt anxious; Happy: I felt happy; Hopeful: I felt hopeful about my future; Look forward: I felt that I had nothing to look forward to; Worth living: I thought my life was not worth living; Good: I felt good about myself; Confide: I felt confident in myself; Unsure: I felt unsure about myself; Failure: I felt like a failure; Able cope: I felt able to cope with my day-to-day life; Unable cope: I felt unable to cope with my day-to-day life; Overwhelm: I felt overwhelmed by my problems or situation; SWEMWBS optimistic: I've been feeling optimistic about the future; SWEMWB relaxed: I've been feeling relaxed. SAFE indicates I felt safe; SWEMWBS, Short Warwick-Edinburgh Mental Well-being Scale.

weighted least squares means and variance adjusted estimation (weighted least squares mean and variance adjusted estimation), just as in the original study. CFA was performed in Mplus (version 8).

## Results

### Sample Characteristics

We recruited a total of 553 respondents, including 96 with lung cancer, 125 with diabetes, 102 with depression, 113 with schizophrenia, 16 with other diseases, and 101 healthy individuals. The face-to-face sample had a mean age of 48 years, with most participants being over 50 years old and having a secondary level of education. Detailed sample characteristics are provided in Table 2. To highlight differences between the face-to-face and online samples, demographic information for the online sample is also presented in Table 3.

### The Model Goodness Index Report

When using the previous online sample to fit the Chinese-adjusted model, the model goodness-of-fit indices use a CFI of 0.908, TLI of 0.901, and RMSEA of 0.071 (90% CI 0.069–0.073). This replicates the original study findings, except for the CFI, which was reported as 0.907 in the original study. In our new face-to-face sample, the model goodness-of-fit indices use a CFI of 0.943, TLI of 0.939, and RMSEA of 0.052 (90% CI 0.050–0.054).

## Discussion

Our findings demonstrate that the goodness-of-fit of the EQ-HWB factor model improves when using a more demographically diverse sample and/or collecting data face-to-face rather than online. In the new Chinese dataset, the RMSEA fell below the 0.06 threshold, indicating improved model fit. Although the CFI and TLI did not reach the conventional 0.95 cutoff, they showed substantial improvement, further supporting the enhanced fit of the model.

One explanation for the better model fit is the increased similarity between the UK and Chinese face-to-face samples compared to the Chinese online sample. For instance, the mean age of the UK sample (49 years) is closer to that of the Chinese face-to-face sample (48 years) than to the Chinese online sample (36 years). In addition, both the UK and Chinese face-to-face samples included patients with diabetes and depression, whereas the Chinese online sample only included patients with depression, making it less comparable. This finding suggests that when generating a factor structure for a QoL questionnaire at the international level and re-testing it at national levels, differences in sample characteristics between nations should be considered. An important takeaway from this study is the necessity of ensuring that data variance is appropriate for its intended analytical purpose.

Despite existing arguments that QoL is a culturally grounded term<sup>20–22</sup> the optimized model fit may illustrate that Chinese respondents show a similar understanding of the items and their relationships, arriving at a conceptual framework of the

**Table 2.** Face-to-face sample demographic.

| Characteristic            | Total N (%) | Lung cancer | Diabetes   | Depression | Schizophrenia | Others    | Healthy   |
|---------------------------|-------------|-------------|------------|------------|---------------|-----------|-----------|
| Subgroup sample size      | 553 (100)   | 96 (17.4)   | 125 (22.6) | 102 (18.4) | 113 (20.4)    | 16 (2.9)  | 101       |
| Gender                    |             |             |            |            |               |           |           |
| Male (1)                  | 307 (55.5)  | 72 (75.0)   | 58 (46.4)  | 41 (40.2)  | 78 (69.0)     | 9 (56.3)  | 49 (48.5) |
| Female (2)                | 246 (44.5)  | 24 (25.0)   | 67 (53.6)  | 61 (59.8)  | 35 (31.0)     | 7 (43.8)  | 52 (51.5) |
| Age group                 |             |             |            |            |               |           |           |
| <18                       | 26 (4.7)    | 0 (0)       | 1 (0.8)    | 25 (24.5)  | 0 (0.0)       | 0 (0.0)   | 0 (0.0)   |
| 18-29                     | 70 (12.7)   | 4 (4.0)     | 2 (1.6)    | 23 (22.5)  | 14 (12.4)     | 2 (12.5)  | 25 (24.8) |
| 30-39                     | 84 (15.2)   | 2 (2.1)     | 6 (4.8)    | 18 (17.6)  | 37 (32.7)     | 0 (0.0)   | 21 (20.8) |
| 40-49                     | 94 (17.0)   | 11 (11.5)   | 13 (10.4)  | 16 (15.7)  | 36 (31.9)     | 0 (0.0)   | 18 (17.8) |
| 50-59                     | 120 (21.7)  | 34 (35.4)   | 32 (25.6)  | 15 (14.7)  | 20 (17.7)     | 3 (18.8)  | 16 (15.8) |
| ≥60                       | 159 (28.8)  | 45 (46.9)   | 71 (56.8)  | 5 (4.9)    | 6 (5.3)       | 11 (68.8) | 21 (20.8) |
| Education                 |             |             |            |            |               |           |           |
| Below primary and primary | 157 (28.4)  | 30 (31.3)   | 20 (16.0)  | 18 (17.6)  | 51 (45.1)     | 7 (43.8)  | 31 (30.7) |
| Secondary to high school  | 305 (55.2)  | 48 (50.0)   | 77 (61.6)  | 57 (55.9)  | 58 (51.3)     | 8 (50.0)  | 57 (56.4) |
| University and above      | 91 (16.5)   | 18 (18.8)   | 28 (22.4)  | 27 (26.5)  | 4 (3.5)       | 1 (6.3)   | 13 (12.9) |

EQ-HWB similar to that of other countries. This highlights the potential of the EQ-HWB for international application and cross-cultural comparisons. The factor structure used in this study does not rule out the possibility of other, potentially better-fitting structures for the Chinese population. However, constructing a more precise factor structure specific to the Chinese context is not our primary goal, because an overly specific structure may hinder the international comparability of the EQ-HWB.

The study provides evidence supporting the international comparability of the EQ-HWB across Eastern and Western cultural contexts. However, it is important to acknowledge that the EQ-HWB is still in development and requires further refinement to become a fully validated instrument. Future studies should validate the EQ-HWB across diverse

populations to assess its performance at both the item and instrument levels, as well as its applicability across different regions. Although cultural factors undoubtedly influence the interpretation and application of the EQ-HWB, the key consideration is whether these differences are substantial enough to distort measurement invariance. In addition, further research should explore how the EQ-HWB can be effectively applied in various practical settings and how it can demonstrate its value in supporting healthcare and policy decision-making.

This study is not without limitations. It remains unclear whether other factors contributed to the differences in model goodness-of-fit indices observed. One possible influence is the mode of data collection—face-to-face versus online. Unlike the online sample, which had no possibility of receiving assistance,

**Table 3.** Online sample demographic.

| Characteristic          | Total n (%) | CHB        | HIV/AIDS  | Depression | GAD       | Healthy    |
|-------------------------|-------------|------------|-----------|------------|-----------|------------|
| Subgroup sample size    | 500 (100)   | 122        | 101       | 107        | 90        | 140        |
| Gender                  |             |            |           |            |           |            |
| Female                  | 300 (60.0)  | 69 (56.6)  | 88 (87.1) | 51 (47.7)  | 48 (53.3) | 72 (51.4)  |
| Age group               |             |            |           |            |           |            |
| 18-29 years of age      | 121 (24.2)  | 17 (13.9)  | 1 (1.0)   | 40 (37.4)  | 21 (23.3) | 61 (43.6)  |
| 30-39                   | 196 (39.2)  | 39 (32.0)  | 50 (49.5) | 37 (34.6)  | 42 (46.7) | 59 (42.1)  |
| 40-49                   | 157 (31.4)  | 62 (50.8)  | 50 (49.5) | 18 (16.8)  | 19 (21.1) | 15 (10.7)  |
| ≥50                     | 26 (5.2)    | 4 (3.3)    | 0 (0)     | 12 (11.2)  | 8 (8.9)   | 5 (3.6)    |
| Education               |             |            |           |            |           |            |
| Secondary education     | 44 (8.8)    | 6 (4.9)    | 4 (4.0)   | 5 (4.7)    | 1 (1.1)   | 20 (14.3)  |
| Undergraduate education | 415 (83.0)  | 107 (87.7) | 97 (96.0) | 87 (81.3)  | 74 (82.8) | 104 (74.3) |
| Postgraduate education  | 41 (8.2)    | 9 (7.4)    | 0 (0)     | 15 (14.0)  | 15 (16.7) | 16 (11.4)  |

CHB indicates Chronic Hepatitis B; GAD, generalized anxiety disorder.



face-to-face respondents had some opportunities to access interviewer support, although it was limited. In addition, face-to-face respondents were allowed to skip certain items, whereas the online sample was required to complete all items without skipping. These differences in data collection methods may have contributed to variations in response patterns and overall data quality, leading to an improvement in model fit. Although the results are promising, further validation across multiple regions in China is warranted to enhance the generalizability and applicability of the EQ-HWB.

## Conclusion

The results of the replication study show that the factor analysis of candidate items of the EQ-HWB in a high-quality Chinese data set is comparable with the factor structure found in other countries.

## Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

## Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.vhri.2025.101501>.

## Article and Author Information

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